Feature Engineering & Impact Analysis

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1. Objective

The goal of this task is to:

- Create **derived features (feature engineering)**
- Measure their **impact on model performance**
- Generate **business insights from transformed data**

2. Dataset (Synthetic E-commerce Dataset)

We created a dataset simulating:

- Product pricing
- Quantity sold
- Discounts
- Marketing spend
- Revenue (target variable)

Why this dataset?

This mimics real-world business scenarios like:

- E-commerce sales analysis
- Marketing ROI optimization
- Pricing strategy evaluation

3. Step-by-Step Theory & Implementation

Step 1: Baseline Model (Before Feature Engineering)

We first train a model using **raw features only**:

Features:

- Price
- Quantity
- Discount
- Marketing Spend

Target:

- Revenue
- This establishes a **benchmark performance**

Step 2: Feature Engineering

We create new features that capture **hidden relationships**:

1. Price After Discount

$$
\text{price_after_discount} = \text{price} \times (1 - \text{discount})
$$

Business meaning: Actual selling price

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2. Revenue per Unit

$$
\text{revenue_per_unit} = \frac{\text{revenue}}{\text{quantity}}
$$

Business meaning: Efficiency of each sale

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3. Marketing Efficiency

$$
\text{marketing_efficiency} = \frac{\text{revenue}}{\text{marketing_spend}}
$$

Business meaning: ROI on marketing spend

Step 3: Model After Feature Engineering

We retrain the model using:

- Original features
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    - Engineered features

Then compare performance

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4. Before vs After Comparison

Model Version	Features Used	Performance (R^2)
Before FE	Raw features only	Lower
After FE	Raw + Engineered	Higher

Feature engineering improves:

- Predictive accuracy
- Interpretability
- Business insight depth

5. Visual (Model Impact)

The notebook includes a **bar chart** comparing:

- Before Feature Engineering
- After Feature Engineering

👉 Clearly shows performance improvement

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6. Business Insights

Key Findings:

1. **Discount-adjusted pricing matters more than raw price**
    - Helps in pricing strategy decisions
2. **Marketing efficiency reveals ROI**
    - Identifies wasteful campaigns
3. **Revenue per unit highlights product performance**
    - Useful for inventory & product strategy